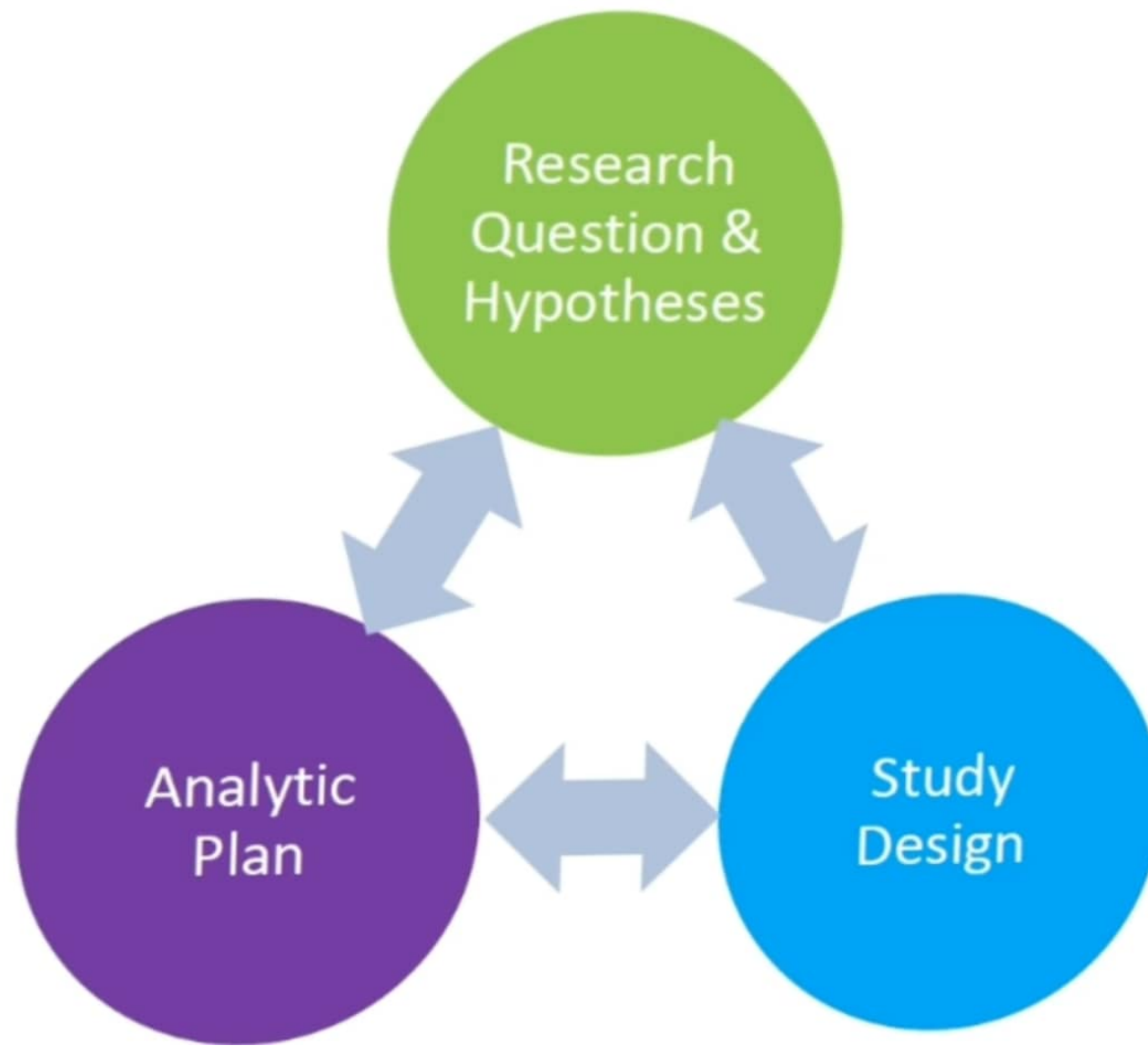
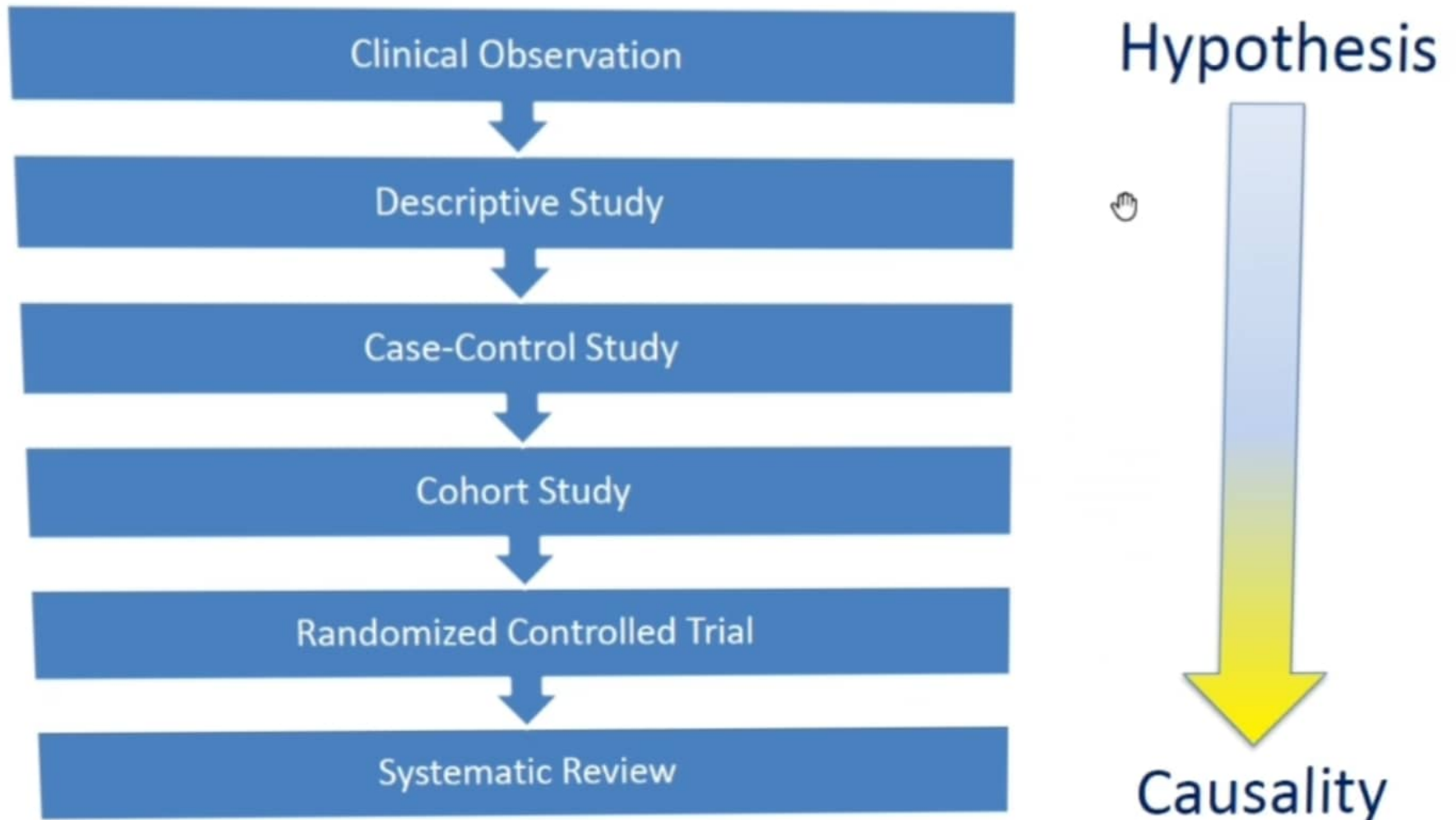


Learning Objectives

- Identify basic epidemiologic study designs and their frequent sequence of study
- Recognize the basic components
- Understand the advantages and disadvantages
- Appropriately select a study design



Basic Study Designs and their Hierarchy





MMWR

1981 June 5;30:250-2

Pneumocystis Pneumonia — Los Angeles

In the period October 1980-May 1981, 5 young men, all active homosexuals, were treated for biopsy-confirmed *Pneumocystis carinii* pneumonia at 3 different hospitals in Los Angeles, California. Two of the patients died. All 5 patients had laboratory-confirmed previous or current cytomegalovirus (CMV) infection and candidal mucosal infection. Case reports of these patients follow.



Adam



Study Design in Epidemiology

- Depends on:
 - **The research question and hypotheses**
 - Resources and time available for the study
 - Type of outcome of interest
 - Type of exposure of interest
 - Ethics



Adam

Study Design in Epidemiology

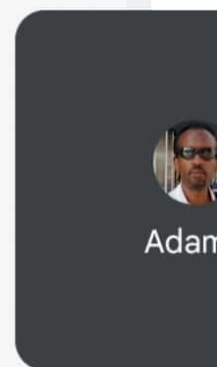
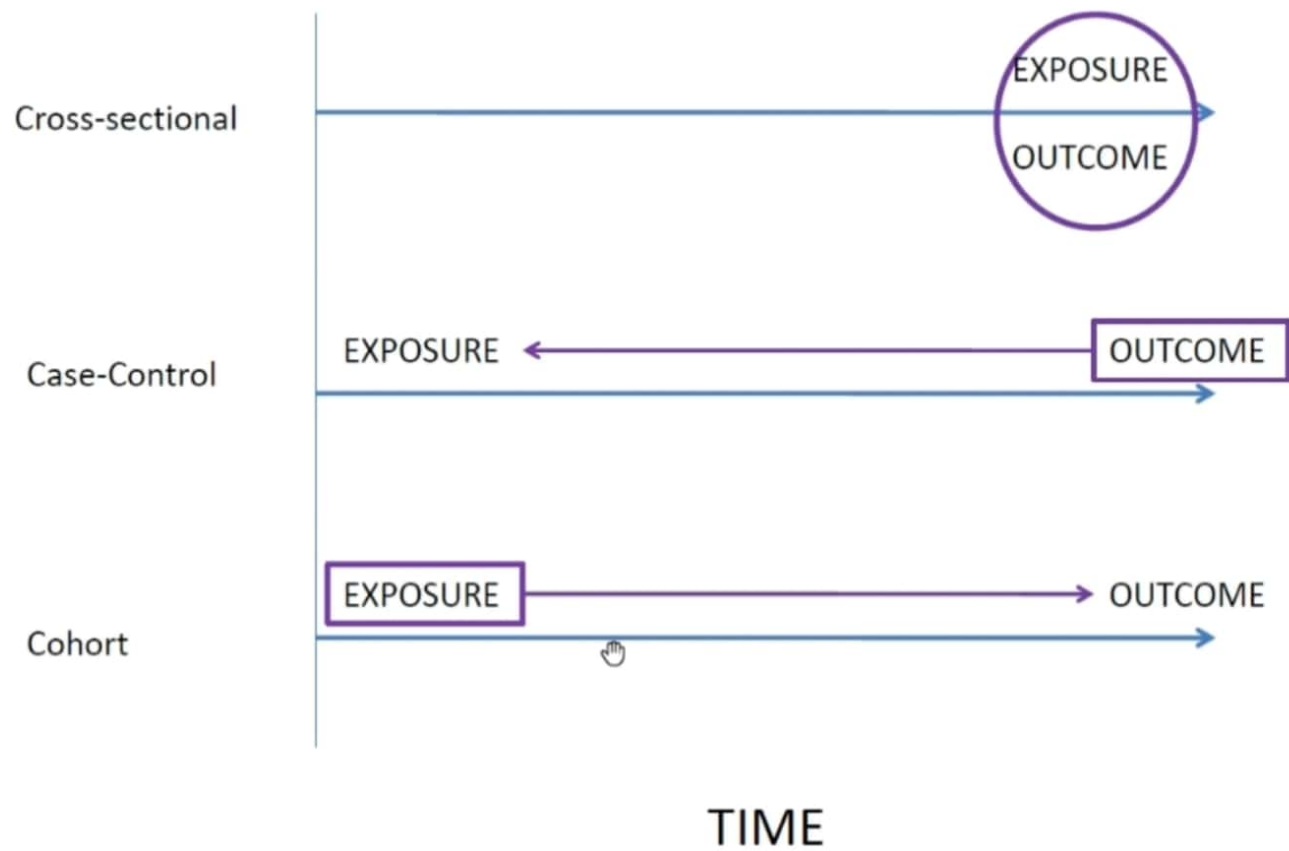
- Includes:
 - The research question and hypotheses
 - Measures and data quality
 - Time
 - Study population
 - Inclusion/exclusion criteria
 - Internal/external validity

Epidemiologic Study Designs

- Descriptive studies
 - Seeks to measure the frequency of disease and/or collect descriptive data on risk factors
- Analytic studies
 - Tests a causal hypothesis about the etiology of disease
- Experimental studies 
 - Compares, for example, treatments



Adam



Cross-sectional studies

- Measure existing disease and current exposure levels at one point in time
- Sample without knowledge of exposure or disease
- Ex. Prevalence studies



Adan

Cross-sectional studies

- Advantages

- Often early study design in a line of investigation
- Good for hypothesis generation
- Relatively easy, quick and inexpensive...*depends on question*
- Examine multiple exposures or outcomes
- Estimate prevalence of disease and exposures

Cross-sectional studies

- Disadvantages
 - Cannot infer causality
 - Prevalent vs. incident disease
 - May miss latent disease
 - May be subject to recall bias



Adam

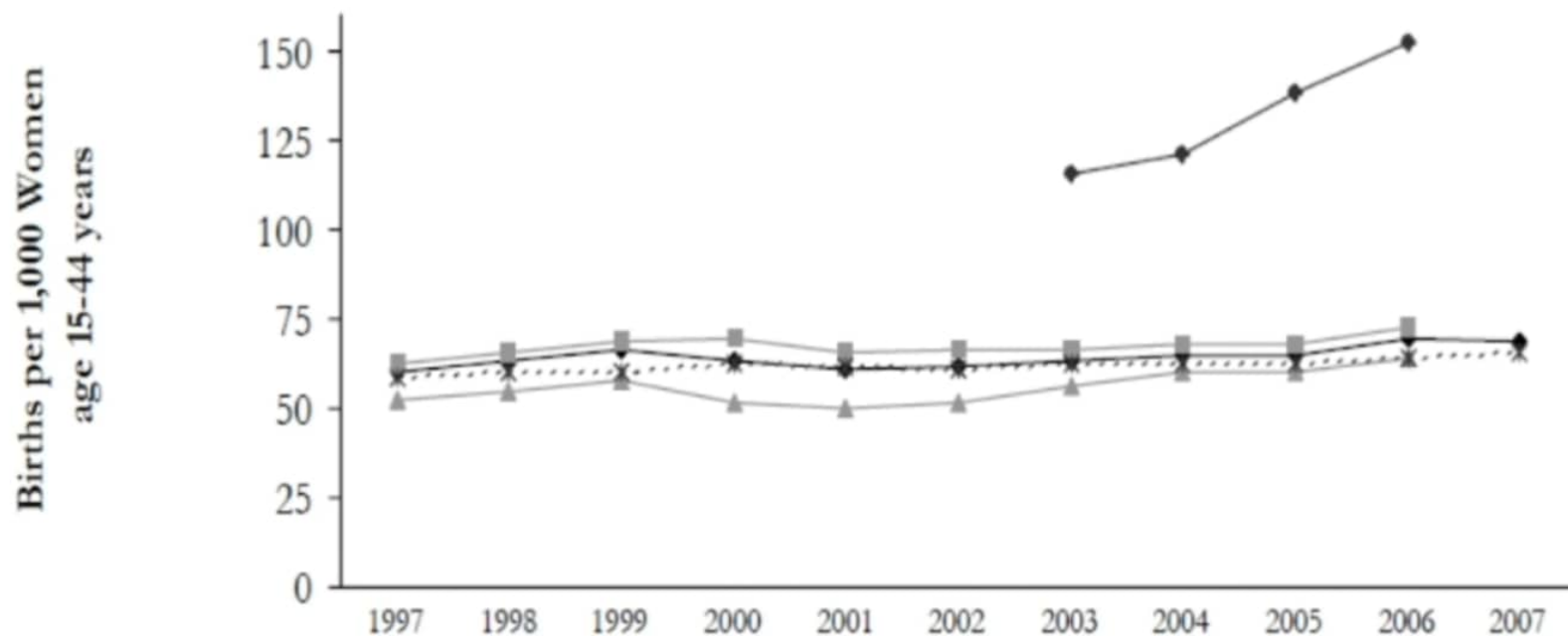
Research Question

- Determine whether there are differences in rates of stroke and myocardial infarction by gender and race among patients.

Hypothesis

- *There will be differences in rates of stroke by gender and race.*
- *There will be differences in rates of myocardial infarction by gender and race.*

General Fertility Rate, Baltimore City by Race and Maryland 1997-2007



◆ Baltimore City All Races	60.0	63.0	66.0	63.1	60.7	61.5	63.0	64.6	65.1	69.3	68.5
■ African American	62.1	65.7	68.4	69.6	65.6	66.3	66.0	67.7	68.2	72.7	
▲ White	52.6	54.9	57.7	51.2	50.3	51.3	55.9	59.9	60.1	64.0	
◆ Hispanic*							115.9	121.0	138.5	152.4	
*** Maryland All Races	58.8	60.0	60.3	62.4	61.5	60.6	62.3	62.2	62.7	64.1	65.5

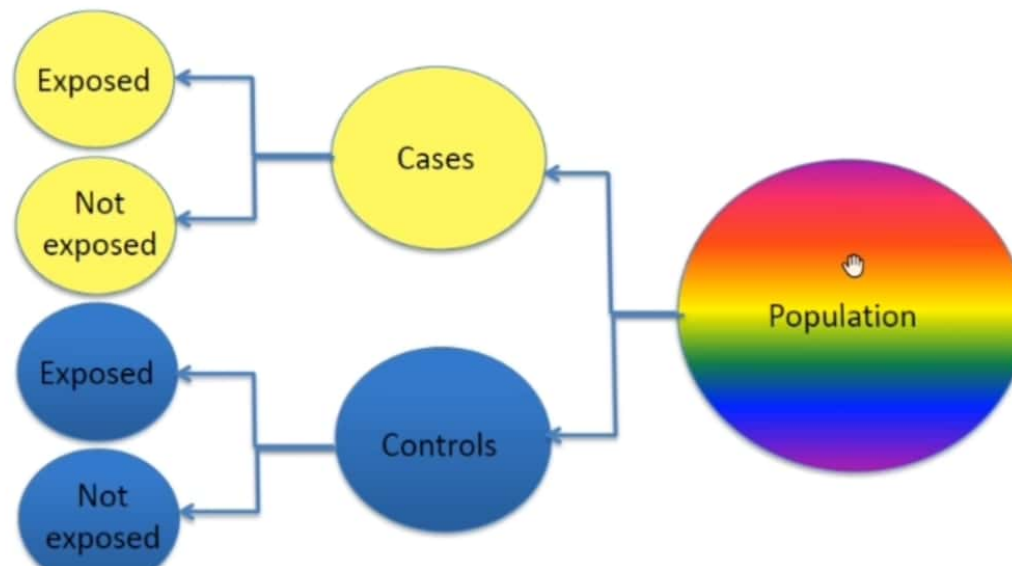


Adam

Case-Control studies

- Identify individuals with existing disease/s and retrospectively measure exposure

← Time



presenting



Case-Control studies

- Advantages

- Good design for rare, chronic and long latency diseases
- Relatively inexpensive (population size and time)
- Allows for the examination of multiple exposures
- Estimate odds ratios
- Hospital-based studies and outbreaks



Case-Control studies

- Disadvantages
 - Multiple outcomes cannot be studied
 - Recall bias
 - Sampling bias
 - Cannot calculate prevalence, incidence, population relative risk or attributable risk
 - Beware of reverse causation

Neonatal Abstinence Syndrome (NAS) and Drug Exposure

Research question

?

Hypothesis 1

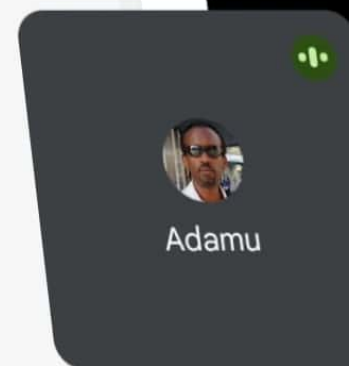
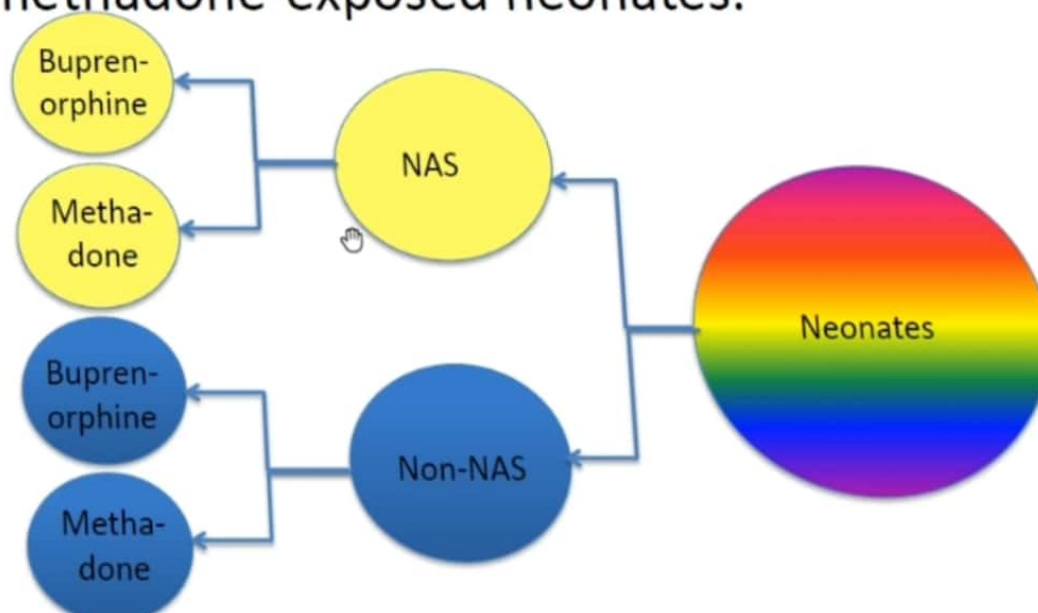
Buprenorphine-exposed neonates will exhibit less NAS than methadone-exposed neonates.



Ada

Case-Control Study Example

- Hypothesis 1: Buprenorphine-exposed neonates will exhibit less NAS than methadone-exposed neonates.



Challenges in Case-Control Studies

- Selection of Controls
 - Sample size
 - Matching (group or individual)
- Selection of Cases
 - Incident or prevalent disease
- Nested case-control study

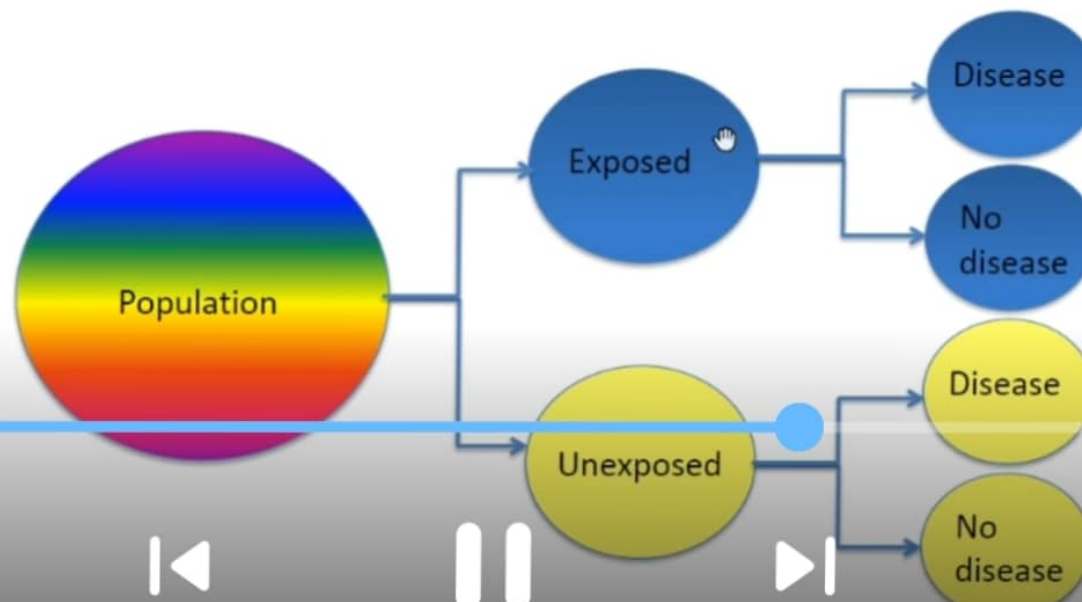


Adamu

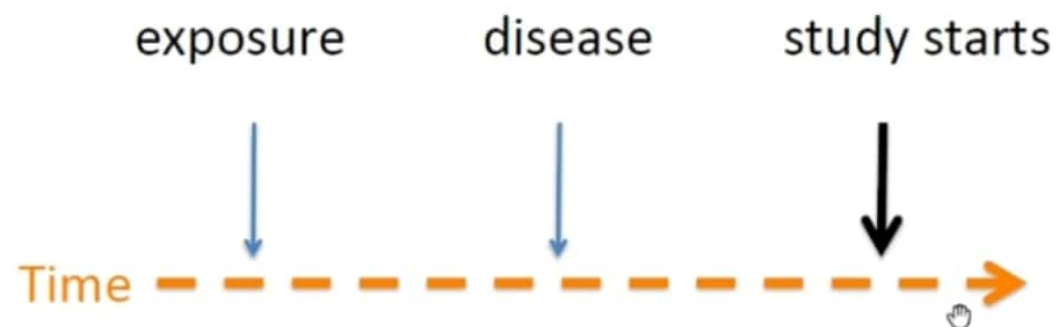
Cohort Studies

- Identify exposed and unexposed individuals and follow them over time measuring outcome/s (Prospective)

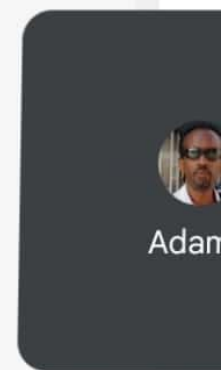
Time — — — — — →



Retrospective Cohort Study



presenting



Cohort Studies

- Advantages

- Measure population-based incidence
- Relative risk and risk ratio estimations
- Rare exposures
- Temporality
- Less likely to be subject to biases (recall and selection as compared to Case-control)
- Possible to assess multiple exposures and/or outcomes



Adamu

Cohort Studies

- Disadvantages

- Impractical for rare diseases and diseases with a long latency
- Expensive
 - Often large study populations
 - Time of follow-up
- Biases
 - Design - sampling, ascertainment and observer
 - Study population – non-response, migration and loss-to-follow-up

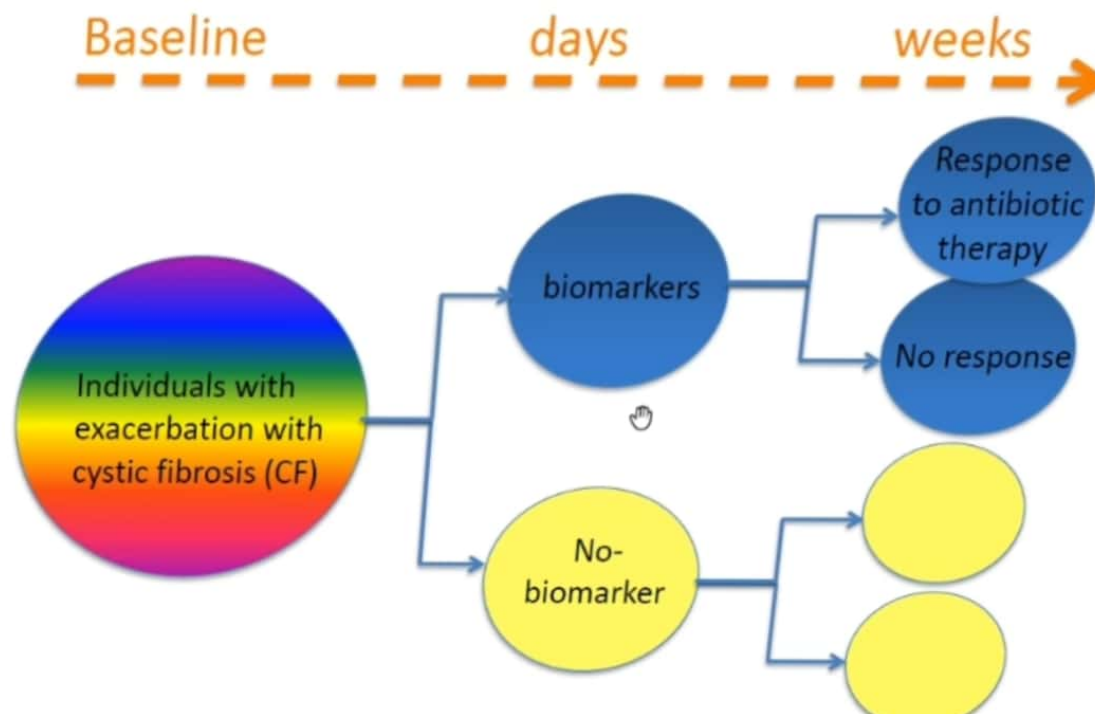
Research Question

Determine whether circulating biomarkers (i.e. C-reactive protein; exhaled breath condensate - pH, hydrogen peroxide, 8-isoprostene, nitrite, nitrate levels; sputum - TNF- α , IL-6, IL-8, IL-1 β , neutrophil elastase; and fractional exhaled nitric oxide) predict individuals who will benefit from initiation of antibiotic therapy for the treatment of a mild decrease in FEV₁.

Hypothesis

Biomarkers at the time of presentation with a mild increase in pulmonary symptoms or small decline in FEV₁ can be used to identify which patients require antibiotics to recover.

Cohort Study



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Important features

- How much selection bias was present?
 - Were only people at risk of the outcome included?
 - Was the exposure clear, specific and measureable?
 - Were the exposed and unexposed similar in all important respects except for the exposure?
- Were steps taken to minimize information bias?
 - Was the outcome clear, specific and measureable?
 - Was the outcome identified in the same way for both groups?
 - Was the determination of the outcome made by an observer blinded to treatment?

Important features

- How complete were the follow-up of both groups?
 - What efforts were made to limit loss to follow-up?
 - Was loss to follow-up similar in both groups?
- Were potential confounding factors sought and controlled for in the study design or analysis?
 - Did the investigators anticipate and gather information on potential confounding factors?
 - What methods were used to assess and control for confounding?

Randomized Controlled Trials (RCTs)

- Experimental: exposure is assigned
- Randomization assignment
 - Random allocation of exposure or treatment
 - Results (or should result!) in two equivalent groups on all measured and unmeasured confounders
- Gold Standard for causal inference

Randomized Controlled Trials

- Advantages

- Least subject to biases of all study designs
(**IF** designed and implemented well...!)

Randomized Controlled Trials

- Disadvantages

- Intent-to-treat
- Loss-to-follow-up
- Randomization issues
- Not all exposures can be “treatments”, i.e. are assignable
- Note: for reporting of RCTs see Altman DG, et al. CONSORT GROUP (Consolidated Standards of Reporting Trials). Ann Intern Med. 2001 Apr 17;134(8):663-94.

Research Question

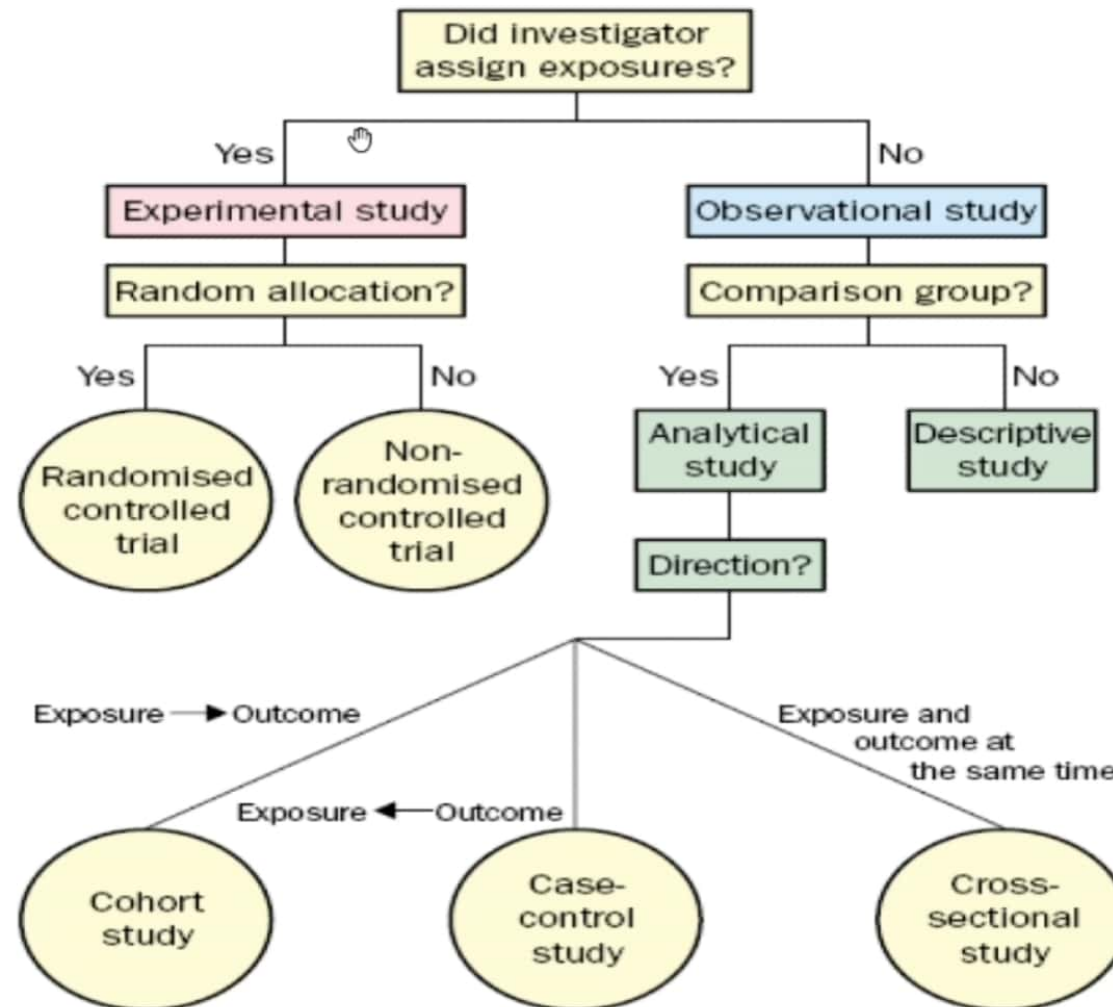
- *To determine whether resident's attitudes and skills in diabetes management and counseling change after a curricular intervention.*
- *To determine whether patient outcomes related to diabetes (i.e. weight, smoking status) change after a curricular intervention among residents.*

Hypothesis

- *Attitudes and skills related to diabetes management and counseling will improve among residents after a curricular intervention.*
- *Fewer patients with diabetes will smoke over time after a curricular intervention among residents.*

Randomization Strategies

- Randomly assigned
- Quasi-randomization
- Block randomization – method of randomization that ensures that at any point in the trial, roughly equal numbers of participants have been allocated to the comparison groups



Study Design

- **Must be defensible**
- **Drives conclusions:**
What do you want to be able to say at the end of the study?